Journal of Biotechnology 164 (2012) 276–280

Contents lists available at SciVerse ScienceDirect

Journal of Biotechnology

journal homepage: www.elsevier.com/locate/jbiotec

Enzymatic synthesis of c-di-GMP using inclusion bodies of *Thermotoga maritima* full-length diguanylate cyclase

A.S. Korovashkina, A.N. Rymko, S.V. Kvach, A.I. Zinchenko*

Institute of Microbiology, National Academy of Sciences, Kuprevich Str. 2, Minsk 220141, Belarus

ARTICLE INFO

Article history:
Received 17 October 2012
Received in revised form 5 December 2012
Accepted 11 December 2012
Available online 26 December 2012

Keywords:
c-di-GMP
Diguanylate cyclase
Enzymatic synthesis
Allosteric site mutation
Inclusion bodies
Thermotoga maritima

ABSTRACT

Recombinant full-length diguanylate cyclases (DGCs) of *Thermotoga maritima* with native and mutant allosteric sites were overexpressed in *Escherichia coli* cells and characterized. It has been shown that target enzymes are produced substantially in the form of active inclusion bodies. Introduction of the mutation in allosteric site resulted in 7-fold increase of the *T. maritima* DGC activity. Possibility of applying full-length DGC of *T. maritima* in the form of inclusion bodies for synthesis of c-di-GMP was originally demonstrated.

© 2012 Elsevier B.V. All rights reserved.

1. Introduction

3′,5′-Cyclic diguanylate (c-di-GMP) is a second messenger molecule found ubiquitously in bacteria (Romling et al., 2005; Jenal and Malone, 2006; Pesavento and Hengge, 2009). In contrast to other second messengers (i.e., cAMP or cGMP), with narrow specialization, c-di-GMP regulates a wide range of molecular processes, including transcription, post-transcriptional control, and the activities of enzymes or large cellular structures (Sudarsan et al., 2008; Tuckerman et al., 2011; Boehm et al., 2010). Physiologically, it affects biofilm formation, cell–cell signaling, differentiation, motility and virulence (Karaolis et al., 2005a; Brouillette et al., 2005; Povolotsky and Hengge, 2012). It was demonstrated that c-di-GMP stimulates the immune system to prevent bacterial infections and is therefore evaluated as a potential vaccine adjuvant (Karaolis et al., 2007; Chen et al., 2010; Burdette et al., 2011). It was also reported that c-di-GMP inhibits cancer cell proliferation in vitro and hence it could find use as an antitumor therapeutic agent (Karaolis et al., 2005b). In bacterial cells c-di-GMP is synthesized by a family of enzymes known as diguanylate cyclases (DGCs) and degraded by c-di-GMP-phosphodiesterases (PDEs). The c-di-GMP signaling is present only in Eukarya and Archea and totally absent in animal species, which makes DGCs interesting targets for antimicrobial or anti-biofilm agents (Antoniani et al., 2010; Sambanthamoorthy et al., 2012).

Characterization of the macromolecules involved in c-di-GMP signaling, including the enzymes and receptors, implies application of c-di-GMP as substrate or ligand for biochemical assay. The c-di-GMP can also serve as starting material for the synthesis of c-di-GMP analogs to be used as chemical probes or inhibitors (Rao et al., 2009). It appears therefore that large quantities of this cyclic dinucleotide are required. Currently c-di-GMP is mainly produced by chemical methods suffering from such limitations as complexity, high cost and low yield of the end product (Yan and Aguilar, 2007; Kiburu et al., 2008). On the other hand, the enzymatic production of c-di-GMP using bacterial diguanylate cyclase domain-containing proteins involves only a single condensation of two GTP molecules (Chan et al., 2004).

Enzymatic large-scale synthesis of c-di-GMP from GTP has been reported for DGC of *Gluconacetobacter xylinum* (Spehr et al., 2011), *Escherichia coli* (Zahringer et al., 2011) and a DGC fragment of *Thermotoga maritima* (Rao et al., 2009). Mutations were introduced in allosteric sites of DGC from *G. xylinum* and *T. maritima* to increase yield of the target product. However, these proteins were produced at relatively low level (10–15 mg/L of culture broth) or required complex and labor-consuming procedure of refolding. Furthermore, the mesophilic proteins of *G. xylinum* and *E. coli* are rather unstable in solution.

Abbreviations: c-di-GMP, 3′,5′-cyclic diguanylate; DGC, diguanylate cyclase; IBs, inclusion bodies; GTP, guanosine-5′-triphosphate; GDP, guanosine-5′-diphosphate; GMP, guanosine-5′-monophosphate; SDS–PAGE, sodium dodecyl sulphate–polyacrylamide gel electrophoresis; PCR, polymerase chain reaction; HPLC, high-performance liquid chromatography.

* Corresponding author. Tel.: +375 172 65 99 68; fax: +375 172 67 47 66.
E-mail address: zinch@mbio.bas-net.by (A.I. Zinchenko).

0168-1656/$ – see front matter © 2012 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.jbiotec.2012.12.006

Full-length DGC of *T. maritima* (GenBank ID: TM1788) comprises catalytic GGDEF domain and hydrophobic domain of unknown function (Rao et al., 2009). It was shown earlier that the synthesis of c-di-GMP from two molecules of GTP requires the DGC protein to function in its dimeric form (Romling et al., 2005). Previously used DGC fragment of *T. maritima* containing only catalytic domain in the reaction mixture corresponds to monomeric form and synthesis of c-di-GMP is achieved via the formation of transient dimer by the protein in the solution (Rao et al., 2009).

Considering the above-mentioned aspects, we presumed that application of full-length DGC from *T. maritima* comprising both catalytic and hydrophobic domains would increase the reaction rate due to formation of stable DGC dimers through hydrophobic interactions. In turn, improving reaction rate will promote launching industrial production of pharmaceutically valuable c-di-GMP.

2. Materials and methods

2.1. Cloning and site-directed mutagenesis

The DGC-encoding gene (GenBank ID: TM1788) was amplified by PCR from the chromosomal DNA of *T. maritima* using the following synthetic oligonucleotide primers: DgcfullF 5′-ATTGTG*CATATG*AAAGTTTCCGGGGGGGAAGT-3′ and DgcfullR 5′-ATCTAGCTCGAGTGAGAGACTGAAATAAGGCAC-3′. *Nde*I (DgcfullF) and *Xho*I (DgcfullR) restriction sites were inserted at 5′ ends of the primers (the bases are in italic). Allosteric center site-directed mutagenesis was performed using megaprimer method (McPherson et al., 2000) with primers: RXXD-f 5′-ACAGGGTT**GCA**AGAAGCGATGTGGTGGC-3′ and RXXD-r 5′-ATCGCTTCT**TGC**AACCCTGTCCAGAATCC-3′ (mutation is in bold). The purified PCR products were digested with *Nde*I and *Xho*I restriction endonucleases and incorporated into the pET24b(+) (Novagen) expression vector, which had been restricted with the same enzymes. Mutation was verified using the BigDye Terminator (version 3.1) Cycle Sequencing Kit on the ABI PRISM 310 Genetic Analyzer (Applied Biosystems, USA). The vectors harboring native and mutant DGC genes fused with a hexahistidine tag at the C-terminus were transformed into *E. coli* strain BL21 (DE3). Cell stocks were stored in 10% glycerol at −80 °C.

2.2. Protein expression and purification

For protein expression, the transformed cells were grown on an orbital shaker set at 37 °C and 250 rpm. Cultivation was performed in 2 L Erlenmeyer flasks containing 500 mL of ZYM505 nutrient medium (Studier, 2005) supplemented with 50 mg/L kanamycin. Production of recombinant enzymes was induced with auto-induction method described by Studier (2005). The cells were then harvested by centrifugation for 10 min at 12,000 × *g*, washed once with buffer A containing 50 mM Tris–HCl (pH 8.0) and 100 mM KCl and slurried in the same buffer. Cells lysis was performed by sonication. Soluble and insoluble fractions were separated by centrifugation for 30 min at 21,000 × *g*. The precipitate containing IBs was washed with 1 M urea solution containing 2% Triton X-100. After centrifugation the precipitate was washed with buffer A to remove urea and Triton X-100 traces and resuspended in the same buffer. The soluble and insoluble fractions were assayed for enzyme activity and protein content.

2.3. Quantification of inclusion bodies concentration

IBs concentration was determined by modified Biuret protein assay. In brief, suspension containing inclusion bodies was boiled during 5 min with 1% SDS. Further measurement was performed according to standard procedure (Dawson et al., 1986).

2.4. Protein content estimation

SDS–PAGE with Coomassie staining was used both to analyze protein composition and to estimate the purity of washed IBs. SDS–PAGE was performed as described by Laemmli (1970) using 12% polyacrylamide gel. Gels were treated according to standard protocols for Coomassie staining, scanned and analyzed with GelAnalyzer 2010a software (http://www.gelanalyzer.com/).

2.5. Enzyme activity assay

DGC activity assay was carried out in 100 μL reaction mixture containing 1 mM GTP, 50 mM Tris–HCl (pH 8.0), 5 mM $MgCl_2$, 250 mM NaCl and 2.5 μM freshly produced DGC. The reaction mixture was incubated at 55 °C and aliquots were taken at different time points. The samples were loaded onto an TSK DEAE-3SW HPLC-column (LKB-Bromma) and eluted at a flow rate 1 mL/min by linear gradient from 3 mM KCl/3 mM KH_2PO_4 (pH 3.3) to 360 mM KCl/360 mM KH_2PO_4 (pH 4.9) for 25 min. The amount of enzyme converting 1 μmol of substrate per minute was assumed as 1 enzyme activity unit.

2.6. c-di-GMP synthesis optimization

Enzyme assays were conducted to determine the optimal pH of the c-di-GMP synthesis in the reaction mixture containing 1 mM GTP, 5 mM $MgCl_2$, 250 mM NaCl, 2.5 μM freshly produced mutant enzyme (DGC^{R158A}) and different buffers: Tris–HCl (pH 7.5–9.0), potassium phosphate buffer (pH 6.0–8.0), glycine–NaOH (9.0–10.6). The reaction mixture was incubated at 55 °C.

To determine the optimal salt concentration reaction was carried out in reaction mixture containing 1 mM GTP, 50 mM Tris–HCl (pH 8.0), 5 mM $MgCl_2$, 2.5 μM freshly produced DGC^{R158A} and 0–500 mM NaCl. The reaction mixture was incubated at 55 °C.

Optimal magnesium ion concentration was determined as described above in the reaction mixture containing different $MgCl_2$ concentrations (0–20 mM).

To determine optimal temperature for c-di-GMP synthesis reaction was carried out at different temperatures (45–65 °C). Aliquots where taken at different time points and analyzed using HPLC as described above.

3. Results and discussion

Large-scale enzymatic synthesis of c-di-GMP using DGC of *G. xylinum* (Spehr et al., 2011), *E. coli* (Zahringer et al., 2011) and DGC fragment of *T. maritima* (Rao et al., 2009) has previously been reported. The most promising from the described enzymes is the fragment of DGC from hyperthermophilic microorganism *T. maritima* due to its high thermostability and product yield. However, this enzyme lacks hydrophobic domain existing in the full-length protein (Rao et al., 2009) which might affect the enzyme activity. We presumed that full-length DGC of *T. maritima* comprising both catalytic and hydrophobic domains would form stable dimers in solution due to hydrophobic interactions and therefore would possess higher enzymatic activity than the truncated enzyme.

Potent product inhibition of DGC by c-di-GMP limiting the yield of target product during enzymatic synthesis is a well-studied phenomenon (Christen et al., 2006; Chan et al., 2004). At the same time, it has been shown that the R158A allosteric site mutation eliminates c-di-GMP feed-back inhibition of truncated DGC of *T. maritima* (Rao et al., 2009). Full-length DGC of *T. maritima* with native allosteric site (DGC^{W}) and DGC carrying allosteric site

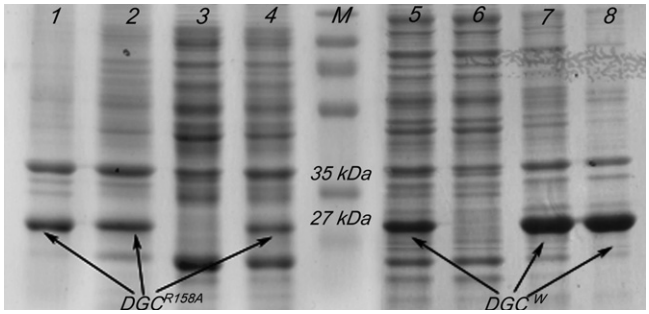


Fig. 1. SDS–PAGE image of crude (lanes 4 and 5), soluble (lanes 3 and 6), insoluble (lanes 2 and 7) and washed with 1 M urea and 2% Triton X-100 insoluble (lanes 1 and 8) cell lysate fractions of *E. coli* pTmadgc (lanes 1–4) and *E. coli* pTmadgc-w7 (lanes 5–8).

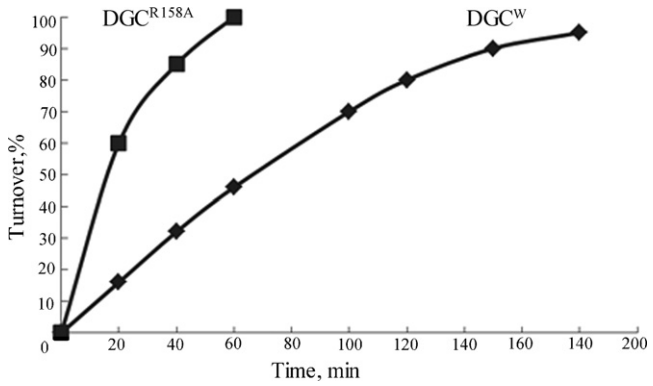


Fig. 2. Kinetics of c-di-GMP synthesis from 1 mM GTP with 2.5 μM DGC^{R158A} and 9.5 μM DGC^{W}.

mutation R158A (DGC^{R158A}) were obtained in the current work to investigate the influence of allosteric inhibition on the activity of full-length enzyme.

Using methods of genetic engineering, two strains producing DGC of *T. maritima* with native (*E. coli* pTmadgc-w7) and mutant (*E. coli* pTmadgc) allosteric site have been constructed. Auto-induction of target proteins was performed in ZYM505 medium. Analysis of protein composition of cell lysate and its soluble and insoluble fractions was conducted with SDS–PAGE (Fig. 1).

Fig. 1 shows that the target proteins are produced mainly in the form of inclusion bodies (IBs). In the soluble fractions the proteins with the molecular weight corresponding to full-length DGC (29 kDa) were not detected. This could be explained by presence of hydrophobic domain at the N-terminus of full-length DGC leading to its aggregation. Expression level of DGC^{R158A} (6%) calculated with program GelAnalyzer 2010a was approximately 4-fold lower than that of DGC^{W} (23%). This might be due to higher enzymatic activity of DGC^{R158A} and its increased toxicity for cells.

An accumulation of insoluble aggregates, widely known as IBs, is a common observation during recombinant protein production in prokaryotic hosts (especially expression of proteins from thermophiles) (Nahalka, 2008). Although IBs have traditionally been described as biologically inert protein clusters, recent findings show that these structures are formed to a considerable extent by functional proteins. IBs are regarded as structurally organized, mechanically stable and biocompatible protein deposits, formed through stereospecific protein–protein cross-molecular interactions (Rodriguez-Carmona et al., 2010).

Taking the above-mentioned aspects into consideration, we analyzed enzymatic activities of both soluble and insoluble fractions of cell lysate. No DGC-activity was revealed in the soluble fractions, while the activity values 4.23 U/g protein for DGC^{R158A} and 1.84 U/g protein for DGC^{W} were recorded in the insoluble fractions. However, GTP was cleaved to GDP and GMP due to presence of interfering phosphatase activity in the enzyme preparations. It required further purification of obtained proteins. For this purpose the precipitate containing IBs was washed with 1 M urea solution containing 2% Triton X-100. Purity of obtained protein preparations was determined using SDS–PAGE (Fig. 1, lanes 1 and 8). The results of performed purification stages are summarized in Table 1.

There was approximately 2.5-fold increase of the total enzymatic activity in the protein preparations after purification procedure. It might be interpreted by presence of proteins with interfering phosphatase activity in the initial cell lysates. In the purified enzyme preparations no phosphatase activity was detected. Purification grade of washed inclusion bodies estimated by GelAnalyzer 2010a software was 31% for DGC^{R158A} and 50% for DGC^{W}. The yield of target proteins was 80 mg/L for DGC^{R158A} and 410 mg/L for DGC^{W}, with specific activities 60.25 and 8.8 U/g of target protein, respectively. Thus, the specific activity of mutant DGC was 7-fold higher than that of native protein. Enzymatic assay showed that, unlike native DGC and truncated DGC, described by Chan et al. (2004) and Rao et al. (2009), both native and mutant enzymes used in the current work were not inhibited by the target product during conversion of 1 mM GTP into c-di-GMP (Fig. 2). We were surprised by the results because the available literature data indicate that DGC is inhibited in presence of low c-di-GMP concentrations, with *Ki* ranging 0.5–44 μM for different enzymes (Chan et al., 2004; Zahringer et al., 2011). We showed that introduction of R158A mutation affected only enzyme activity rather than feed-back product inhibition.

Taking into account the obtained findings, DGC^{R158A} was chosen for further investigation. Systemic screening of pH, temperature, salt and magnesium ions concentrations was carried out to optimize conditions of the reaction catalyzed by the DGC^{R158A}. The enzyme was found to be the most efficient at pH 10.0 and at relatively high salt concentrations (300–500 mM NaCl) (Fig. 3A and B). Variation of the salt (LiCl, NaCl, KCl) had only marginal effect (data not shown). It was shown that magnesium ions were crucial for DGC^{R158A} activity, with an optimum at 2.5 mM. Higher concentrations resulted in slight inhibition of the enzymatic activity (Fig. 3C).

Table 1
Results of DGC^{R158A} and DGC^{W} purification.

Fraction	DGC^{R158A}					DGC^{W}				
	Total activity, U	Total protein, g	Specific activity, U/g protein	Relative purification factor	Recovery, %	Total activity, U	Total protein, g	Specific activity, U/g protein	Relative purification factor	Recovery, %
Crude cell lysate	2	1.27	1.57	1	100	1.4	1.9	0.74	1	100
Precipitate of the cell lysate	2.3	0.55	4.23	2.7	115	1.9	1.032	1.84	2.5	135
Precipitate of the cell lysate after washing with 1 M urea + 2% Triton X-100	4.8	0.26	18.68	12	240	3.6	0.825	4.4	5.9	259

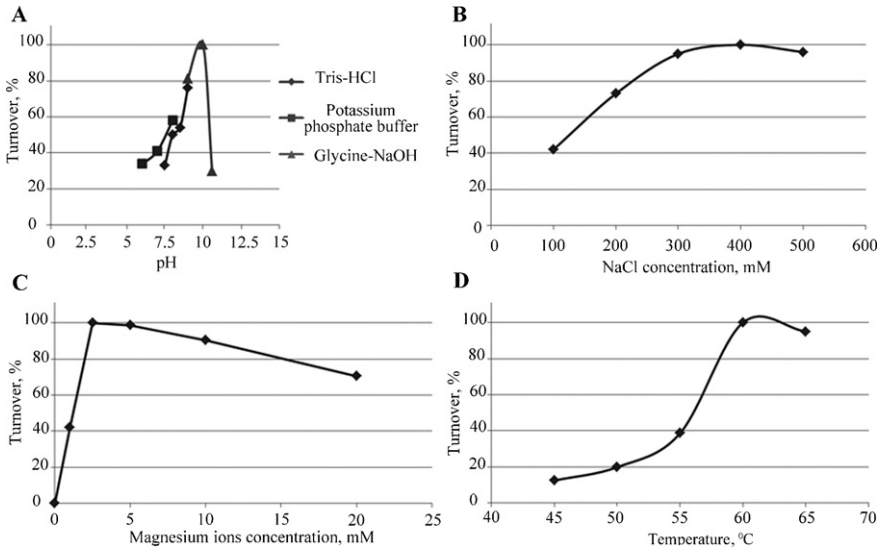


Fig. 3. Optimization of c-di-GMP synthesis conditions. Effect of pH (A), salt (B), magnesium ion concentration (C) and temperature (D) on enzymatic activity of DGC^{R158A}. Assay conditions are described in Section 2.

The temperature 60 °C proved the optimal value for c-di-GMP synthesis (Fig. 3D).

Enzyme activity was measured under optimized reaction conditions: $MgCl_2$–2.5 mM, NaCl–300 mM, glycine–NaOH (pH 10.0)–50 mM, GTP–1 mM and 2.5 μM DGC^{R158A}. Reaction mixture was incubated at 60 °C. Specific activity of DGC^{R158A} was 151.5 U/g of target protein, i.e. 2.5-fold higher than the activity estimated under non-optimized conditions.

Analytical synthesis of c-di-GMP was performed in 1 ml reaction mixture using 2.5 μM DGC^{R158A} and 1 mM GTP. About 100% of GTP was converted into c-di-GMP in 1 h under the above-described conditions. The identity of the obtained product to commercial standard (Biolog) was confirmed by HPLC and UV detection at different pH values.

4. Conclusion

This work resulted in construction of strains *E. coli* pTmadgc-w7 and *E. coli* pTmadgc producing full-length recombinant DGC of *T. maritima* with native and mutant allosteric sites. It has been shown that the target enzymes are produced substantially in the form of active inclusion bodies, which facilitates the protein purification process and subsequent isolation of the enzyme from the reaction mixture. 7-Fold increase in activity of the *T. maritima* DGC was triggered by mutation introduced into allosteric site.

Acknowledgement

This work was supported by the Belarus State Research Program "Fundamental Basics of Biotechnologies" (Grant 3.07).

References

Antoniani, D., Bocci, P., Maciąg, A., Raffaelli, N., Landini, P., 2010. Monitoring of diguanylate cyclase activity and of cyclic-di-GMP biosynthesis by whole-cell assays suitable for high-throughput screening of biofilm inhibitors. Applied Microbiology and Biotechnology 85, 1095–1104.

Boehm, A., Kaiser, M., Li, H., Spangler, C., Kasper, C.A., Ackermann, M., Kaever, V., Sourjik, V., Roth, V., Jenal, U., 2010. Second messenger-mediated adjustment of bacterial swimming velocity. Cell 141, 107–116.

Brouillette, E., Hyodo, M., Hayakawa, Y., Karaolis, D.K., Malouin, F., 2005. 3′,5′-Cyclic diguanylic acid reduces the virulence of biofilm-forming Staphylococcus aureus strains in a mouse model of mastitis infection. Antimicrobial Agents and Chemotherapy 9, 3109–3113.

Burdette, D.L., Monroe, K.M., Sotelo-Troha, K., Iwig, J.S., Eckert, B., Hyodo, M., Hayakawa, Y., Vance, R.E., 2011. STING is a direct innate immune sensor of cyclic di-GMP. Nature 478, 515–519.

Chan, C., Paul, R., Samoray, D., Amiot, N.C., Giese, B., Jenal, U., Schirmer, T., 2004. Structural basis of activity and allosteric control of diguanylate cyclase. Proceedings of the National Academy of Sciences of the United States of America 101, 17084–17089.

Chen, W., KuoLee, R., Yan, H., 2010. The potential of 3′,5′-cyclic diguanylic acid (c-di-GMP) as an effective vaccine adjuvant. Vaccine 28, 3080–3085.

Christen, B., Christen, M., Paul, R., Schmid, F., Folcher, M., Jenoe, P., Meuwly, M., Jenal, U., 2006. Allosteric control of cyclic di-GMP signaling. Journal of Biological Chemistry 281, 32015–32024.

Dawson, R.M.C., Elliott, D.C., Elliott, W.H., Jones, K.M., 1986. Data for Biological Research, 3rd ed. Clarendon Press, Oxford.

McPherson, M.J., Moller, S.G., Beynon, R., Howe, C., 2000. PCR: The Basics from Background to Bench. BIOS Scientific Publishing, New York, USA.

Jenal, U., Malone, J., 2006. Mechanisms of cyclic-di-GMP signaling in bacteria. Annual Review of Genetics 40, 385–407.

Karaolis, D.K.R., Rashid, M.H., Chythanya, R., Luo, W., Hyodo, M., Hayakawa, Y., 2005a. C-di-GMP (3′-5′-cyclic diguanylic acid) inhibits *Staphylococcus aureus* cell–cell interactions and biofilm formation. Antimicrobial Agents and Chemotherapy 49, 1029–1038.

Karaolis, D.K.R., Cheng, K., Lipsky, M., Elnabawi, A., Catalano, J., Hyodo, M., 2005b. 3′,5′-Cyclic diguanylic acid (c-di-GMP) inhibits basal and growth factor-stimulated human colon cancer cell proliferation. Biochemical and Biophysical Research Communications 329, 40–45.

Karaolis, D.K.R., Means, T.K., Yang, D., Takahashi, M., Yoshimura, T., Muraille, E., Philpott, D., Schroeder, J.T., Hyodo, M., Hayakawa, Y., Talbot, B.G., Brouilette, E., Malouin, F., 2007. Bacterial c-di-GMP is an immunostimulatory molecule. Journal of Immunology 178, 2171–2181.

Kiburu, I., Shurer, A., Yan, L., Sintim, H.O., 2008. A simple solid-phase synthesis of the ubiquitous bacterial signaling molecule, c-di-GMP and analogues. Molecular BioSystems 4, 518–520.

Laemmli, U.K., 1970. Cleavage of structural proteins during the assembly of the heat of bacteriophage T4. Nature 227, 680–685.

Nahalka, J., 2008. Physiological aggregation of maltodextrin phosphorylase from *Pyrococcus furiosus* and its application in a process of batch starch degradation to α-D-glucose-1-phosphate. Journal of Industrial Microbiology and Biotechnology 35, 219–223.

Pesavento, C., Hengge, R., 2009. Bacterial nucleotide-based second messengers. Current Opinion in Microbiology 12, 170–176.

Povolotsky, T.L., Hengge, R., 2012. 'Life-style' control networks in *Escherichia coli*: signaling by the second messenger c-di-GMP. Journal of Biotechnology 160, 10–16.

Rao, F., Pasunooti, S., Ng, Y., Zhuo, W., Lim, L., Liu, A.W., Liang, Z.X., 2009. Enzymatic synthesis of c-di-GMP using a thermophilic diguanylate cyclase. Analytical Biochemistry 389, 138–142.

Rodriguez-Carmona, E., Cano-Garrido, O., Seras-Franzoso, J., Villaverde, A., Garcia-Fruitos, E., 2010. Isolation of cell-free bacterial inclusion bodies. Microbial Cell Factories 9, 71.

Romling, U., Gomelsky, M., Galperin, M.Y., 2005. C-di-GMP: the dawning of a novel bacterial signalling system. Molecular Microbiology 57, 629–639.

Sambanthamoorthy, K., Sloup, R.E., Parashar, V., Smith, J.M., Kim, E.E., Semmelhack, M.F., Neiditch, M.B., Waters, C.M., 2012. Identification of small molecules that

antagonize diguanylate cyclase enzymes to inhibit biofilm formation. Antimicrobial Agents and Chemotherapy 56, 5202–5211.
Spehr, V., Warrass, R., Hocherl, K., Ilg, T., 2011. Large-scale production of the immunomodulator c-di-GMP from GMP and ATP by an enzymatic cascade. Applied Biochemistry and Biotechnology 165, 761–775.
Studier, F.W., 2005. Protein production by auto-induction in high-density shaking cultures. Protein Expression and Purification 41, 207–234.
Sudarsan, N., Lee, E.R., Weinberg, Z., Moy, R.H., Kim, J.N., Link, K.H., Breaker, R.R., 2008. Riboswitches in eubacteria sense the second messenger cyclic Di-GMP. Science 321, 411–413.
Tuckerman, J.R., Gonzalez, G., Gilles-Gonzalez, M.A., 2011. Cyclic di-GMP activation of polynucleotide phosphorylase signal-dependent RNA processing. Journal of Molecular Biology 407, 633–639.
Yan, H., Aguilar, A.L., 2007. Synthesis of 3′,5′-cyclic diguanylic acid (cdiGMP) using 1-(4-chlorophenyl)-4-ethoxypiperidin-4-yl as a protecting group for 2′-hydroxy functions of ribonucleosides. Nucleosides, Nucleotides and Nucleic Acids 26, 189–204.
Zahringer, F., Massa, C., Schirmer, T., 2011. Efficient enzymatic production of the bacterial second messenger c-di-GMP by the diguanylate cyclase YdeH from *E. coli*. Applied Biochemistry and Biotechnology 163, 71–79.